

Stationary distribution

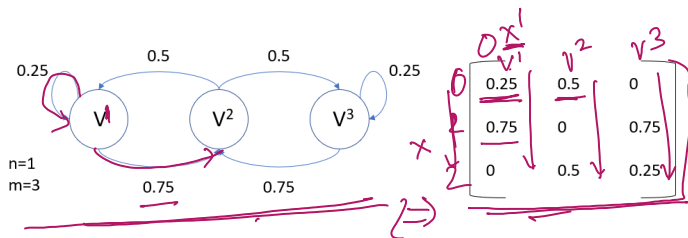
We are interested in $T(x|x')$ for which as $t \rightarrow \infty$

$$\pi(x) = \sum_{x'} \pi(x') T(x|x')$$

has a unique solution for a given $T(x|x')$ and $\pi(x)$ should be reachable from any initial state x^0 via Markov walks using $T(x|x')$

→ For any $T(x|x')$ there always exists a solution to $\pi(x)$, but there might be multiple solutions, or MCMC may not converge to the unique solution.

Example: computing stationary distribution



$$\pi(x) = [\pi_1, \pi_2, \pi_3]$$

$$\pi(x) = \sum_{x'} \pi(x') T(x|x')$$

$$\pi_1 = 0.25\pi_1 + 0.5\pi_2$$

$$\pi_2 = 0.75\pi_1 + 0.75\pi_3$$

$$\pi_3 = 0.5\pi_2 + 0.25\pi_3$$

$$\pi_1 + \pi_2 + \pi_3 = 1$$

$$\pi_1 = \pi_3 = 2/7, \pi_2 = 3/7$$

$$\pi = [2/7, 3/7, 2/7]$$

Example - with multiple solutions

There might exist $T(x|x')$ for which $\pi(x) = \sum_{x'} \pi(x') T(x|x')$ has multiple solutions

$$T(x|x') = 1 \text{ if } x = x'$$

$$T(x|x') = 0 \text{ otherwise}$$



$\pi(x)$ can be any distribution

Gives an Identity matrix

Hence Infinite # of solutions

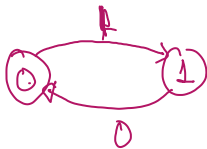
$$\underline{P_t(x)} = \underline{\delta(x^0)}$$

Example: Stationary distribution unreachable via Markov walks

$$\underline{P_t(x^t)} \approx \underline{P_{t+1}(x^{t+1})}$$

Even when $T(x|x')$ has a unique stationary distribution $\pi_T(x)$, this distribution $\pi_T(x)$ may not be the one that we can reach or converge to via MCMC walks.

$$T(x|x') = 1 \text{ if } x \neq x'$$
$$T(x|x') = 0 \text{ otherwise}$$



$$\pi_T(0) = \pi_T(1) = 1/2 \leftarrow \text{unique}$$

$$t=0 \quad x^0 = v_1$$

$$t=1 \quad x^1 = v_2$$

for any general t , $P_t(x^t = 1) = v_1$ if t is even and $P_t(x^t = 1) = v_2$ if t is odd

$P_t(x)$ is not getting close to $P_{t+1}(x)$. Thus there is no *convergence* to the stationary distribution.

Ergodicity

Definition: Markov chains with a unique $\pi(x)$ that can be reached via MCMC steps irrespective of starting point are called ergodic.

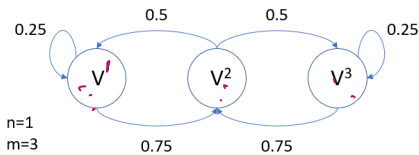
Theorem

When number of states is finite, then a MC is ergodic iff it is regular

Definition: A MC is regular if $\exists k$ such that any x to x' can be reached in exactly k steps.

$$P(X_k=i | X_0=j) = T^k(i,j)$$

Examples of Ergodicity



0.25	0.5	0
0.75	0	0.75
0	0.5	0.25

$$k = 1 \quad \times$$

$$k = 2 \quad \checkmark$$

$$\pi = \left[\frac{2}{7}, \frac{3}{7}, \frac{2}{7} \right]$$

Simpler Sufficient condition for Ergodicity^{di}_n

Lemma

A finite state MC is regular when [not iff]:

- 1 every state has a self loop with non-zero probability
- 2 between any two states $\exists$ a path of non-zero probability

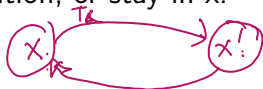
Metropolis Hastings Sampling: A broad class of Markov chains KF 12.3.4:

How to design a transition distribution $T(x \rightarrow x')$ for an arbitrary high-dimensional distribution?

- Metropolis Hastings offers a general recipe that is applicable for any distribution for which it is easy to compute ratio of probabilities $P(X)/P(X')$.
- Useful for distributions like undirected graphical models where the normalizer is intractable.
- Also, for distributions defined via energy functions.

Metropolis Hastings MC Sampling from $P(x)$

- 1 Choose any proposal distribution for transferring from x to x'
 $T^Q(x \rightarrow x')$
- 2 Use T^Q to propose a transition from x to x' . We accept the proposal with probability $A(x \rightarrow x')$ and transition, or stay in x .
- 3 Where

$$\underline{A(x \rightarrow x')} = \min \left[1, \frac{P(x') T^Q(x' \rightarrow x)}{P(x) T^Q(x \rightarrow x')} \right] \text{ satisfies this}$$


The overall transition function is:

$$\rightarrow T(x \rightarrow x') = \underline{T^Q(x \rightarrow x')} \underline{A(x \rightarrow x')} \quad x \neq x'$$

$$\rightarrow T(x \rightarrow x) = \underline{T^Q(x \rightarrow x)} + \sum_{x' \neq x} \underline{T^Q(x \rightarrow x')} \underline{(1 - A(x \rightarrow x'))}$$

Property of MH sampling

Theorem

If $T(x|x')$ is regular, then $P(x)$ is a stationary distribution of the MCMC.

We will prove shortly using another property that the MH sampling satisfies.

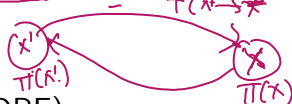
An implication of this theorem is that, we can start from any proposed transition distribution that is regular, and using only the acceptance probability correction, achieve the target distribution. The calculation of $A()$ only depends on the ratio of probabilities, so the normalizer cancels out.

Proof: Reversible Chains

Definition: A finite state Markov chain T is reversible if $\exists$ a unique π such that $\forall x, x' \in \mathcal{X}$

$$\pi(x') T(x' \rightarrow x) = \pi(x) T(x \rightarrow x')$$

Above is called the Detailed balance Equation (DBE)



Theorem

If $\pi(x)$ satisfies above then $\pi(x)$ is a stationary distribution of T .

Proof

$$\rightarrow \pi(x) = \sum_{x'} \pi(x') T(x' \rightarrow x)$$

$$\begin{aligned} \sum_{x'} \pi(x') T(x' \rightarrow x) &= \sum_{x'} \pi(x) T(x \rightarrow x') \\ &= \pi(x) \sum_{x'} T(x \rightarrow x') \\ &= \pi(x) \end{aligned}$$

This implies that $\pi(x)$ is a stationary distribution of T

Proof (continued)

The A chosen as

$$A(x \rightarrow x') = \min \left[1, \frac{\pi(x') T^Q(x' \rightarrow x)}{\pi(x) T^Q(x \rightarrow x')} \right] \text{ satisfies this}$$

satisfies the detailed balance equation for $x \neq x'$

$$\rightarrow \pi(x) T^Q(x \rightarrow x') \underline{A(x \rightarrow x')} = \pi(x') T^Q(x' \rightarrow x) \underline{A(x' \rightarrow x)}$$

[Proof easy: Either numerator $\pi(x') T^Q(x' \rightarrow x)$ is lower or the denominator. Assume numerator. Then

$A(x \rightarrow x') < 1, A(x' \rightarrow x) = 1$. Plug in these values and DBE will be satisfied.]

Continued.

Example from book

Let us say we desire a stationary distribution:

$$\pi = [\pi_1, \pi_2, \pi_3] = [2/Z, 3/Z, 2/Z]$$

Earlier we had started with a T that gave this π . Now we choose an arbitrary T^Q and compute A .

① Example $T^Q(x \rightarrow x') = 1/3$

② Compute $A(1 \rightarrow 2) = \min[1, \frac{\pi(2)T^Q(1|2)}{\pi(1)T^Q(2|1)}] = \min(1, 3/2) = 1$

③ Compute $A(2 \rightarrow 3) = 2/3$

$$\min\left[1, \frac{\pi(3)}{\pi(2)}\right] = \frac{2}{3}$$

Designing a proposal transition function

- 1 Random $T_Q(\mathbf{x}|\mathbf{x}') \sim \mathbf{x}' + N(0, \sigma I)$

$$\mathbf{x} \in \mathbb{R}^d$$

[https://chi-feng.github.io/mcmc-demo/app.html?
algorithm=RandomWalkMH&target=banana](https://chi-feng.github.io/mcmc-demo/app.html?algorithm=RandomWalkMH&target=banana)

- 2 Gibbs Sampling
- 3 Langevin Monte Carlo

Gibbs Sampling

$$x = [x_1, x_2, \dots, x_n]$$

Gibbs Sampling samples from one state to another that differs at most in one position. It first samples a dimension to change (say i), and then samples a value of the dimension based on the conditional probability $P(x_i | x_{-i})$.

The corresponding $T(x|x')$ matrix then turns out to be:

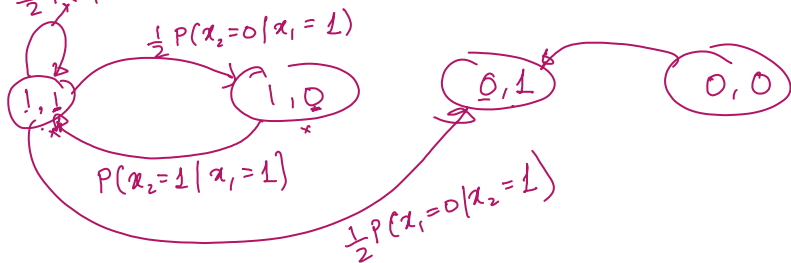
$$T(x|x') = \begin{cases} 0 & \text{if } x \text{ and } x' \text{ differ in more than one } i \\ \frac{1}{n} P(x_i | x'_{-i}) & \text{if } x \neq x' \text{ but } x_{-i} = x'_{-i} \\ \frac{1}{n} \sum_{i=1}^n P(x_i | x_{-i}) & \text{if } x = x' \end{cases}$$

Handwritten notes:
 $x' = [0, 0, 0]$ with indices 1, 2, 3 below the zeros.
 $n=3$
 $i=2$
 $P(x_2 | x_1=0, x_3=0)$
 $x_{-2} = x'_{-2}$

Example: $n=2, m=2, x_i \in \{1, 2\}$

Gibbs sampling example and demo

$$\mathcal{X} = \{ (1, 1), (1, 2), (2, 1), (2, 2) \}$$
$$\frac{1}{2} P(x_1=0|x_2=0) + \frac{1}{2} P(x_2=0|x_1=0)$$



[https://chi-feng.github.io/mcmc-demo/app.html?](https://chi-feng.github.io/mcmc-demo/app.html?algorithm=GibbsSampling&target=banana)
algorithm=GibbsSampling&target=banana

Gibbs sampling and Ergodicity

MC chain defined by Gibbs sampling is ergodic and has stationary distribution $P(\mathbf{x} = x_1 \dots, x_n)$ when $P(\mathbf{x})$ is positive

Proof:

- 1 Self loop probability $> 0 \forall \mathbf{x}$ $T(\mathbf{x}|\mathbf{x}') > 0$ if $P(\mathbf{x})$ is a positive distribution
- 2 $\mathbf{x}' \rightarrow \mathbf{x}$ can be reached with non-zero probability
- 3 We will next show that $\pi(\mathbf{x}) = P(\mathbf{x})$ when $T(\mathbf{x}|\mathbf{x}')$ using the detailed balance equation.

Reversibility check for Gibbs

To show that: $P(\mathbf{x}) T(\mathbf{x}'|\mathbf{x}) = P(\mathbf{x}') T(\mathbf{x}|\mathbf{x}')$, $\mathbf{x} \neq \mathbf{x}'$

Proof: $\mathbf{x}$ and $\mathbf{x}'$ can only differ in one position. Let that be i . Then :

$\mathbf{x}' = [x'_i, \mathbf{x}_{-i}]$ and $\mathbf{x} = [x_i, \mathbf{x}_{-i}]$

$$\begin{aligned} P(\mathbf{x}) T(\mathbf{x}'|\mathbf{x}) &= \frac{1}{n} P(\mathbf{x}) P(x'_i|\mathbf{x}_{-i}) \\ &= \frac{1}{n} P(\mathbf{x}) \frac{P(x'_i, \mathbf{x}_{-i})}{\sum_{\mathbf{x}} P(x_i = x, \mathbf{x}_{-i})} \\ &= \frac{1}{n} P(\mathbf{x}) \frac{P(\mathbf{x}')}{\sum_{\mathbf{x}} P(x_i = x, \mathbf{x}_{-i})} \\ &= \frac{1}{n} P(\mathbf{x}') \frac{P(x_i, \mathbf{x}_{-i})}{\sum_{\mathbf{x}} P(x_i = x, \mathbf{x}_{-i})} = P(\mathbf{x}') T(\mathbf{x}|\mathbf{x}') \end{aligned}$$

This gives us a proof that $P(\mathbf{x})$ is a stationary distribution under Gibbs sampling.

Calculating $P(x_i | \mathbf{x}_{-i})$ for undirected graphical models I

$$\begin{aligned}
 P(x_1, x_2, \dots, x_n) &= \frac{1}{Z} \prod_c \psi_c(\mathbf{x}_c) \\
 P(x_i | \mathbf{x}_{-i}) &= \frac{P(x_1, x_2, \dots, x_n)}{\sum_{x_i} P(x_1, x_2, \dots, x_n)} = \frac{\frac{1}{Z} \prod_c \psi_c(\mathbf{x}_c)}{\frac{1}{Z} \sum_{x_i} \prod_c \psi_c(\mathbf{x}_c)} = \frac{\prod_{c: i \in C} \psi_c(\mathbf{x}_c)}{\sum_{x_i} \prod_{c: i \in C} \psi_c(\mathbf{x}_c)}
 \end{aligned}$$

Handwritten notes in the image include:

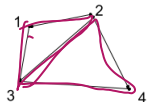
- A pink box around $\prod_c \psi_c(\mathbf{x}_c)$ in the first equation.
- A pink arrow pointing from the boxed term to the numerator of the second equation.
- Handwritten annotations in the second equation:
 - A pink arrow pointing to $P(x_i | \mathbf{x}_{-i})$.
 - A pink arrow pointing to $\sum_{x_i}$ in the denominator.
 - Handwritten cancellation of terms: $\frac{\prod_c \psi_c(\mathbf{x}_c)}{\sum_{x_i} \prod_c \psi_c(\mathbf{x}_c)} = \frac{\prod_{c: i \in C} \psi_c(\mathbf{x}_c)}{\sum_{x_i} \prod_{c: i \in C} \psi_c(\mathbf{x}_c)}$, with terms like $\frac{\pi_{c: i \notin C}}{\sum_{x_i} \pi_{c: i \notin C}}$ being crossed out.

In the last step we have used the fact that potentials that do not contain the variable i can be taken outside the summation in the denominator, and they cancel out. This shows that to calculate the conditional probability we only need to consider clique potentials where the i -th variable appears. More importantly, in the

Calculating $P(x_i | \mathbf{x}_{-i})$ for undirected graphical models II

denominator we are only summing over values of one variable. This is quite tractable for discrete variables.

Example



$$\begin{aligned}
 P(x_1 | \mathbf{x}_{-1}) &= \frac{\psi_{123}(x_1, x_2, x_3) \psi_{234}(x_2, x_3, x_4)}{\sum_{x_1} \psi_{123}(x_1, x_2, x_3) \psi_{234}(x_2, x_3, x_4)} \\
 &= \frac{\psi_{123}(x_1, x_2, x_3)}{\sum_{x_1} \psi_{123}(x_1, x_2, x_3)}
 \end{aligned}$$

Limitations of Gibbs

- ① For continuous distribution $P(x_i|x_{-i})$ may not be easy to obtain.
- ② When variables are highly correlated Gibbs sampling which allows only local moves might have high probability states whose neighbours are mostly low probability leading to poor mixing.

Example of poor mixing I

$$\epsilon = 0.001$$

$$\begin{aligned} P(\underline{x_1}, \underline{x_2}) &= \frac{1 - \epsilon}{2} \text{ if } \underline{x_1} = \underline{x_2} \quad x_i \in \{0, 1\} \\ &= \frac{\epsilon}{2} \quad \text{if } \underline{x_1} \neq \underline{x_2} \quad \epsilon \text{ is small} \end{aligned}$$

Assume ϵ is small. The above then is an example of a distribution where there are two modes, (0,0) and (1,1) of equal probability. The other two assignments (0,1) and (1,0) have very low probability. An ideal sampler should generate (0,0) and (1,1) with equal probability, irrespective of whether you start from (0,0) or (1,1). We will show that if we restrict to Gibbs sampling, the samples generated will be strongly biased by the starting point.

Limitations of Gibbs

Applying the formula for Gibbs transition we will get that

$$\frac{1}{n} \sum_{i=1}^n P(x_i | X_{-i})$$

$$T([0,0]||[0,0]) = \frac{1}{2} [P(x_1 = 0 | x_2 = 0) + P(x_2 = 0 | x_1 = 0)]$$

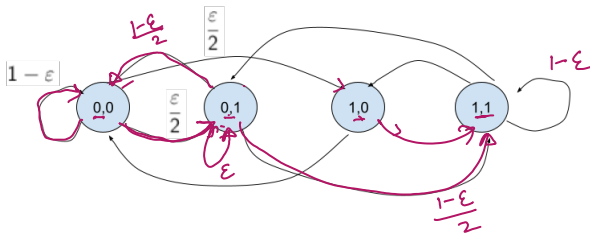
$$= \frac{1}{2} \left[\frac{P(0,0)}{P(0,0) + P(1,0)} + \frac{P(0,0)}{P(0,0) + P(0,1)} \right]$$

$$= \frac{1}{2} \left[\frac{(1-\epsilon)/2}{(\frac{1-\epsilon}{2}) + \frac{\epsilon}{2}} + \frac{(1-\epsilon)/2}{(\frac{1-\epsilon}{2}) + \frac{\epsilon}{2}} \right] = 1 - \epsilon$$

Show that: $T([0,0]||[0,1]) = (1 - \epsilon)/2$ Show that:
 $T([0,1]||[0,0]) = \epsilon/2$

Limitations of Gibbs

Overall Markov Chain transitions are

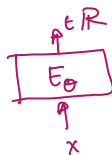


Langevin Monte-Carlo

(https://en.wikipedia.org/wiki/Metropolis-adjusted_Langevin_algorithm)

Sampling from an arbitrary differentiable (in $\mathbf{x}$) function eg. Neural network representing $P(\mathbf{x})$ [eg: audio, images]

$$P(\mathbf{x}) \propto \exp(-E_{\theta}(\mathbf{x})) \quad P(\mathbf{x}) = \frac{\exp(-E_{\theta}(\mathbf{x}))}{Z_{\theta}}$$



where $E_{\theta}(\mathbf{x}) \mapsto R$ is an arbitrary differentiable function in $\mathbf{x}$ and Z_{θ} is intractable to compute

But given two $\mathbf{x}$ and $\mathbf{x}'$ easy to compute the ratio of their probabilities as $\frac{P(\mathbf{x})}{P(\mathbf{x}')} = \frac{\exp(-E_{\theta}(\mathbf{x}))}{\exp(-E_{\theta}(\mathbf{x}'))}$.

How to design $T^Q(\mathbf{x}'|\mathbf{x})$?